

# Tutorial 1: Number Systems

## AMA3707 Real Analysis

The definition references below are imported from the lecture notes. Click a reference to open the corresponding definition in `lecture01.pdf`.

**Question 1.** Using definition 1.1.1, determine which of

$$-3, \quad 0, \quad 1, \quad 7, \quad \frac{3}{2}$$

are natural numbers.

**Question 2.** Refer to definition 1.1.2. Decide whether each of

$$-5, \quad 0, \quad 2.4, \quad \sqrt{9}$$

is an integer, and briefly justify each answer.

**Question 3.** Using definition 1.1.3, prove that 0.125 is rational by writing it in the form  $p/q$ , where  $p, q \in \mathbb{Z}$  and  $q \neq 0$ .

**Question 4.** Use definitions 1.1.2 and 1.1.3 to explain why every integer is also a rational number.